Jocelyn Rego

MACHINE LEARNING RESEARCH SCIENTIST

Los Angeles, CA

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Machine learning research scientist building intelligent systems across applied domains. PhD bridging neuroscience and machine learning for robust and efficient systems. Driven by curiosity and a commitment to building reliable, rigorous artificial intelligence to solve real-world scientific problems.

Education

Drexel University *Philadelphia, PA*

PHD COMPUTER SCIENCE *2019 - 2024*

- Thesis: *"Learning from the Brain: Leveraging Biology for Robust and Efficient Machine Intelligence"*
- Candidacy: *"Sight Unseen: Generative Models and Robust Computer Vision"*
- Advisor: Dr. Edward Kim
- Research Interests: Biologically Inspired Machine Learning, Energy-Based Models, Sparse Coding, Neuromorphic Computing,Computational Neuroscience, Cognitive Science

MS COMPUTER SCIENCE *2019 - 2021*

- Summa Cum Laude (GPA: 3.96)
- Upsilon Pi Epsilon, Honor Society for Computing and Information Disciplines

Villanova University *Villanova, PA*

BS COGNITIVE AND BEHAVIORAL NEUROSCIENCE *2015 - 2019*

- Concentration in Cognitive Science, Minors in Computer Science and Psychology
- Magna Cum Laude (GPA: 3.87)
- Psi Chi, Psychology and Neuroscience Honors Society

Research Experience

Intelligent Systems Center - HRL Laboratories, LLC *Malibu, CA*

SCIENTIST IV *2025 - Present*

SCIENTIST III *2022 - 2025*

- Developed real-time ML systems for manufacturing quality assurance, applying neural networks and gradient-boosting classifiers to sensor data for inline defect detection.
- Built ML components for an automated circuit layout system, combining graph neural networks for component placement with LLM-based generation for design-rule-compliant layouts.
- Designed a multimodal verification and validation pipeline for safety-critical AI systems, using conformal inference to detect distribution shift at test time with statistical guarantees.
- Developed closed-loop systems that model cognitive states from behavioral and physiological signals in real time, adapting system behavior to improve human-machine team performance across operational tasks.
- Additional experience includes large-scale behavioral dataset construction for team performance modeling and robotics orchestration with ROS2.

SPARSE Coding Lab - Drexel University *Philadelphia, PA*

GRADUATE RESEARCH ASSISTANT *2019 - 2024*

- Developed a novel energy-based architecture combining sparse coding via the locally competitive algorithm with Hopfield dynamics and equilibrium propagation, unifying robust image classification, dictionary learning, and local learning in a single biologically plausible framework.
- Built hierarchical sparse coding models of biological vision with lateral competition and top-down feedback, demonstrating natural adversarial robustness, strong neural response prediction, and perceptual selectivity across multiple levels of visual processing.

Computer and Computational Sciences Division - Los Alamos National Lab *Los Alamos, NM*

GRADUATE RESEARCH ASSISTANT, ISTI STUDENT FELLOW *2021 - 2022*

- Developed novel spiking neuron models for unsupervised dictionary learning from images and video, targeting deployment on neuromorphic hardware.
- Designed a biologically plausible multiscale competitive feedback mechanism modeling perceptual selectivity in visual processing for robust computer vision.

Programming Languages & Tools

Python (PyTorch, XGBoost, scikit-learn, HuggingFace Transformers, and standard scientific stack), LaTeX, Git

Select Publications

E. A. Cranford, C. Lebiere, D. Morrison, T. H. J. Kim, **J. Rego**, B. Marsh, M. Levy, et al. "Cognitive Models of Reverse Engineering Processes", Cognitive Systems Research, 2026.

Y. Esfandiari, **J. Rego**, A. Meyer, J. Gallagher, M. Levy. "LLMs for Analog Circuit Design Continuum ", arXiv preprint, 2025.

D. Warmsley, K. Choudhary, **J. Rego**, E. Viani, P. K. Pilly. "Self-Assessment in Machines Boosts Human Trust", Frontiers in Robotics and AI, 2025.

E. Kim, M. Daniali, **J. Rego**, G. T. Kenyon. "The Selectivity and Competition of the Mind's Eye in Visual Perception", International Conference on Acoustics, Speech and Signal Processing, ICASSP 2024.

J. Rego, Y. Watkins, G. T. Kenyon, E. Kim, M. Teti. "A Novel Model of Primary Visual Cortex Based on Biologically Plausible Sparse Coding", Application of Machine Learning, SPIE 2023.

S. V. Dibbo, S. Mansingh, **J. Rego**, G. T. Kenyon, J. Moore, M. Teti. "How Can Neuroscience Help Us Build More Robust Deep Neural Networks?" ICML 2nd Workshop on New Frontiers in Adversarial Machine Learning, 2023.

S. Nesbit, A. O'Brien, **J. Rego**, G. Parpart, C. Gonzalez, G. T. Kenyon, E. Kim, T. C. Stewart, Y. Watkins. "Think Fast: Time Control in Varying Paradigms of Spiking Neural Networks", International Conference on Neuromorphic Systems, ICONS 2022.

G. Parpart, C. Gonzalez, T. C. Stewart, E. Kim, **J. Rego**, A. O'Brien, S. Nesbit, G. T. Kenyon, Y. Watkins. "Dictionary Learning with Accumulator Neurons", International Conference on Neuromorphic Systems, ICONS 2022.

J. Carter, **J. Rego**, D. Schwartz, V. Bhandawat, E. Kim. "Learning Spiking Neural Network Models of Drosophila Olfaction", International Conference on Neuromorphic Systems, ICONS 2020.

E. Kim, **J. Rego**, Y. Watkins, G. T. Kenyon. "Modeling Biological Immunity to Adversarial Examples", IEEE International Conference on Computer Vision and Pattern Recognition, CVPR 2020.

Teaching Experience

Machine Learning (CS613), Teaching Assistant — *Drexel University*
Applications of Machine Learning (CS614), Teaching Assistant
Game AI (CS611), Teaching Assistant

Fellowships & Awards

2023 **Dean's Student Leadership and Service Award**, Drexel University, CCI
2021 **Information Science and Technology Institute Fellowship**, Los Alamos National Lab
2020 **Graduate Cohort for Women: Travel Award**, Computing Research Association - WP
2015-2019 **Villanova Scholarship**, Villanova University

Organizations & Leadership

2022-2023 **CCI Doctoral Student Association: Vice President**, Drexel University
2019-2022 **Women in Computing Society**, Drexel University
2019-2022 **Artificial Intelligence Club**, Drexel University
2019 **Board of Academic Integrity**, Villanova University

Additional Professional Experience

2018-2019 **Undergraduate Research Assistant**, Word Recognition and Auditory Perception Lab, Villanova University
2016-2019 **Undergraduate Research Assistant**, Temporal Perception Lab, Villanova University
2017-2018 **Instructional Technology Intern**, Villanova Technology Services
2017 **Healthcare Informatics Intern**, New York City Health + Hospitals